Radicals in iterated quadratic *abc*-transfers

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Abstract

Let (a_0, b_0, c_0) be a primitive positive triple with $a_0 + b_0 = c_0$, where a_0, b_0 have opposite parity, and iterate the classical quadratic transfer

$$(a, b, a+b) \longmapsto \bigl(4ab, (a-b)^2, (a+b)^2\bigr).$$

Writing $d_j = a_j - b_j$, $R_n = \operatorname{rad}(a_n b_n c_n)$, and

$$W_n = \prod_{j<n} \frac{|d_j|}{\operatorname{rad}(|d_j|)},$$

we prove an exact radical identity. If $\cos\theta = (b_0 - a_0)/c_0$, then

$$\frac{R_n}{c_n} = \frac{R_0}{c_0}\,\frac{|\sin(2^n\theta)|}{2^n \sin\theta\, W_n}.$$

One explicit Baker–Wüstholz estimate gives

$$\log(c_n/R_n) = \log W_n + \Theta_{a_0,b_0}(n), \qquad \log c_n = 2^n \log c_0.$$

Thus every such orbit eventually consists of *abc*-hits, while a fixed asymptotic quality gap is equivalent to positive-power growth of W_n. The orbit is also a dyadic subfamily of known negative-discriminant Lucas *abc*-triples. Our orbit-adapted argument recovers along dyadic indices an archimedean limit already implied by general all-index Lucas estimates. For the seed $(1, 8, 9)$,

$$|d_j| = \frac{1}{2}\bigl|V_{2^{j+1}}(2, 9)\bigr|.$$

Standard Lucas valuation theory turns W_n into an aggregate of Lucas–Wieferich contributions. A reproducible search found no square prime divisor for $p \le 10^7$ and $0 \le j \le 50$; the computation is not used in the proofs.

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1 Introduction

For a positive integer m, write $\operatorname{rad}(m) = \prod_{p|m} p$. A primitive triple of positive integers (a, b, c) with $a + b = c$ is called an *abc*-hit if $\operatorname{rad}(abc) < c$. Its quality is

$$q(a, b, c) = \frac{\log c}{\log \operatorname{rad}(abc)}.$$

The abc conjecture predicts that, for every $\varepsilon > 0$, only finitely many such triples have $q(a,b,c) > 1+\varepsilon$; see Granville and Tucker [6] for an introduction.

The identity

$$4ab + (a-b)^2 = (a+b)^2 \tag{1.1}$$

is a classical polynomial transfer for abc-triples. It appears in Oesterlé's discussion of the abc–Szpiro connection [12, p. 170] and in the catalogue of Martin and Miao [10, §2.4]; see also van der Horst [19, §2.3]. Iterating this exact transfer to obtain infinitely many hits, together with one-step quality control, was described by Holsztyński [9] and Guninski [7]. Thus neither the transfer nor its iteration is new. For a primitive triple with odd c, the exact one-step formula

$$\operatorname{rad}\big((b-a)^2(4ab)c^2\big) = \operatorname{rad}(|b-a|)\operatorname{rad}(abc)$$

follows immediately from the disjoint prime supports proved in Lemma 1 (take $n=1$); no external result is needed for this specialization. Martin and Miao [10, §3.1] iterate a different transfer and obtain one-sided radical and quality bounds. Classical families already include $(1, 9^k-1, 9^k)$ [1], so neither the growth law $c_{n+1} = c_n^2$ nor the mere production of infinitely many hits is a novelty claim here.

Our purpose is quantitative. We first prove an exact disjoint prime-support factorization along the whole orbit. It separates the radical loss into an archimedean product of Chebyshev values and a nonarchimedean product W_n containing every repeated prime factor. The first product telescopes to one sine value. A single linear-form estimate then bounds its logarithm above and below on the linear scale n, exponentially smaller than $\log c_n$. This gives an exact radical identity, effective two-sided control, and a two-way equivalence between the asymptotic abc-quality of the orbit and aggregate repeated-prime growth.

These orbits also sit inside a Lucas-sequence construction already studied by Bolvardizadeh [4, Ch. 2]. Her treatment of quality assumes positive discriminant, and her list of future directions isolates a missing archimedean limit when the discriminant is negative [4, pp. 63–64]. We identify our orbit as a constrained dyadic subfamily of that negative-discriminant setting and give a direct, seed-specific estimate along dyadic indices. The later general theorem of Hajdu and Tijdeman [8, Theorem 2.1 and Remark 1] gives non-real Lucas growth estimates for all indices, and therefore also implies this limit in greater generality. Our linear-form argument is included because it is shorter in the fixed-seed orbit and interfaces directly with the radical identity. Related dyadic Lucas factorizations and square-free kernels were studied by Ribenboim [14].

For the illustrative seed $(1,8,9)$, the nonarchimedean terms belong to the known sequence

$$3^m T_m(1/3) = \frac{1}{2} V_m(2,9),$$

whose terms and Chebyshev–Lucas description are recorded as OEIS A025172 [11]. Pairwise coprimality at dyadic indices, rank congruences, and the individual Lucas–Wieferich criterion are standard consequences of Lucas-sequence theory [13, 18]. The orbit identity, its telescoped effective estimate, and the resulting aggregate quality equivalence are the contribution of this note. A bounded search of the mathematical literature and citation trails through July 26, 2026 found no earlier source containing these whole-orbit statements; this is not a claim that the search was exhaustive.

The conclusions do not prove or disprove the abc conjecture.

2 A general quadratic orbit

Let

$$0 < a_0 < b_0, \qquad a_0 + b_0 = c_0, \qquad \gcd(a_0, b_0) = 1, \tag{2.1}$$

and suppose that a_0, b_0 have opposite parity. This ordering loses no generality because the transfer is symmetric in a_0, b_0. Define

$$(a_{n+1}, b_{n+1}, c_{n+1}) = \big(4a_n b_n, (a_n - b_n)^2, c_n^2\big), \qquad d_n = a_n - b_n. \tag{2.2}$$

Identity (1.1) gives $a_n + b_n = c_n$, and

$$c_n = c_0^{2^n}, \qquad d_{n+1} = c_n^2 - 2d_n^2. \tag{2.3}$$

Lemma 1 (Prime supports)**.** *Every triple in (2.2) is primitive. The integers d_j are odd, pairwise coprime, and coprime to $a_0b_0c_0$. Consequently, if $R_n = \mathrm{rad}(a_n b_n c_n)$, then*

$$R_n = R_0 \prod_{j=0}^{n-1} \mathrm{rad}(|d_j|). \tag{2.4}$$

Proof. Opposite parity is preserved by (2.2), so c_n, d_n are odd. At the seed, $\gcd(c_0, d_0) = \gcd(a_0 + b_0, a_0 - b_0) = 1$, since this gcd divides $2\gcd(a_0, b_0)$ and both arguments are odd. If $\gcd(c_n, d_n) = 1$, then

$$\gcd(c_{n+1}, d_{n+1}) = \gcd(c_n^2, c_n^2 - 2d_n^2) = \gcd(c_n^2, 2d_n^2) = 1,$$

because c_n is odd. Thus $\gcd(c_n, d_n) = 1$ for every n. Any common divisor of a_n, b_n divides both c_n and d_n, so every triple is primitive.

We first exclude the primes of the seed. Let p be an odd prime. If $p \mid c_0$, then $p \mid c_k$ for every k, whereas $p \nmid d_0$; (2.3) shows inductively that $p \nmid d_k$. If $p \mid a_0$, then $d_0 \equiv -c_0 \pmod p$, and the congruence $d_k \equiv -c_k \pmod p$ propagates under (2.3). If $p \mid b_0$, then $d_0 \equiv c_0 \pmod p$, after which $d_1 \equiv -c_1 \pmod p$ and the same invariant propagates. Thus no prime of $a_0b_0c_0$ divides a d_j.

If $p \mid d_m$, then $p \nmid c_m$, and (2.3) gives

$$d_{m+1} \equiv c_{m+1} \pmod p, \qquad d_{m+2} \equiv -c_{m+2} \pmod p.$$

The latter congruence persists, so p divides no later d_k. This proves pairwise coprimality.

For $n \geq 1$, $b_n = d_{n-1}^2$, while induction in (2.2) shows that the prime support of a_n is the support of a_0b_0, together with the supports of $d_0, \ldots, d_{n-2}$. All these supports and the support of c_0 are disjoint. Equation (2.4) follows. $\square$

Set

$$x_0 = \frac{b_0 - a_0}{c_0} = \cos\theta, \qquad 0 < \theta < \frac{\pi}{2}.$$

Normalizing (2.3) gives

$$-\frac{d_n}{c_n} = T_{2^n}(x_0) = \cos(2^n\theta). \tag{2.5}$$

The complex number

$$z = e^{i\theta} = \frac{b_0 - a_0 + 2\sqrt{-a_0b_0}}{c_0} \tag{2.6}$$

is not a root of unity. Indeed, if it were, then $z+z^{-1}=2(b_0-a_0)/c_0$ would be a rational algebraic integer. Since $\gcd(c_0,b_0-a_0)=1$ and c_0 is odd and greater than 1, this is impossible. In particular, none of the sine or cosine values used below vanishes.

Define

$$t_j=\frac{|d_j|}{c_j}=|\cos(2^j\theta)|,\qquad W_n=\prod_{j=0}^{n-1}\operatorname{cosoc}(|d_j|),\qquad \operatorname{cosoc}(m)=\frac{m}{\operatorname{rad}(m)}. \tag{2.7}$$

Theorem 2 (Exact orbit identity)**.** *For every $n\geq 0$,*

$$\boxed{\frac{R_n}{c_n}=\frac{R_0}{c_0}\,\frac{|\sin(2^n\theta)|}{2^n\sin\theta\,W_n}.} \tag{2.8}$$

Consequently,

$$\frac{c_n}{R_n}\geq\frac{2^{n+1}\sqrt{a_0b_0}}{R_0}=\frac{2\sqrt{a_0b_0}}{R_0\log c_0}\log c_n. \tag{2.9}$$

Thus every orbit satisfying (2.1) eventually consists of abc-hits. If the seed is already a hit, then every member of the orbit is a hit.

Proof. Since

$$\prod_{j<n}c_j=c_0^{2^n-1}=\frac{c_n}{c_0},$$

equations (2.4) and (2.7) first give

$$\frac{R_n}{c_n}=\frac{R_0}{c_0}\frac{\prod_{j<n}t_j}{W_n}. \tag{2.10}$$

Iterating $\sin(2x)=2\sin x\cos x$ gives the exact telescope

$$\prod_{j=0}^{n-1}|\cos(2^j\theta)|=\frac{|\sin(2^n\theta)|}{2^n\sin\theta}, \tag{2.11}$$

which proves (2.8). Now use $W_n\geq 1$, $|\sin(2^n\theta)|\leq 1$, and $\sin\theta=2\sqrt{a_0b_0}/c_0$ to obtain (2.9). If $R_0<c_0$, (2.10) and $0<t_j<1$ prove the final assertion. $\square$

Remark 3. The lower bound (2.9) has order only $\log c_n$. Much larger unconditional excesses $c/\operatorname{rad}(abc)$ are known in specially designed families. In particular, Bright [5] proves that

$$c>\operatorname{rad}(abc)\exp\left(\frac{6.563\sqrt{\log c}}{\log\log c}\right)$$

for infinitely many primitive triples; see also [6, 1]. The point of (2.8) is its exactness and its validity for every primitive opposite-parity seed.

3 Effective archimedean control and quality

We record a completely effective estimate for the only oscillating factor in (2.8). Let

$$C_{\mathrm{BW}}=18\cdot 3!\cdot 2^3\cdot 64^4\log 8,\qquad H_0=\max\left\{\frac{1}{2}\log c_0,\frac{\theta}{2},\frac{1}{2}\right\},\qquad \kappa=C_{\mathrm{BW}}H_0\frac{\pi}{2}. \tag{3.1}$$

Theorem 4 (One-shot Baker estimate)**.** *For $n \geq 2$,*

$$|\sin(2^n\theta)| > \frac{2}{\pi}\, 2^{-\kappa(n+1)}. \tag{3.2}$$

If

$$F_n = \log\frac{c_n}{R_n} - \log W_n, \qquad C_0 = \log\frac{c_0}{R_0} + \log\sin\theta,$$

then the lower bound below holds for every $n \geq 0$, and the upper bound holds for $n \geq 2$:

$$C_0 + n\log 2 \leq F_n < C_0 + \log\frac{\pi}{2} + \kappa\log 2 + (1+\kappa)n\log 2. \tag{3.3}$$

In particular,

$$\log\frac{c_n}{R_n} = \log W_n + \Theta_{a_0,b_0}(n). \tag{3.4}$$

Proof. Put $N = 2^n$, choose an integer k nearest to $N\theta/\pi$, and take the principal logarithms $\log z = i\theta$ and $\log(-1) = i\pi$. Then

$$\Lambda_N = N\log z - k\log(-1) = i(N\theta - k\pi), \qquad |\Lambda_N| \leq \pi/2. \tag{3.5}$$

The form is nonzero because z is not a root of unity. Moreover $|k| \leq N$.

We apply the explicit linear-form estimate of Baker and Wüstholz [2, Theorem 1]. The field $\mathbb{Q}(z,-1) = \mathbb{Q}(\sqrt{-a_0b_0})$ has degree $D = 2$. The primitive minimal polynomial

$$c_0X^2 - 2(b_0 - a_0)X + c_0$$

has both roots on the unit circle, so $h(z) = \frac{1}{2}\log c_0$. The modified heights in the theorem may therefore be taken as H_0 and $\pi/2$. For $n \geq 2$, take the strict coefficient bound $B = 2N \geq 8$. The theorem gives

$$\log|\Lambda_N| > -\kappa\log(2N). \tag{3.6}$$

Since

$$|\sin(N\theta)| = |\sin(N\theta - k\pi)| \geq \frac{2}{\pi}|\Lambda_N|,$$

equation (3.2) follows. Finally, taking logarithms in (2.8) gives the exact expression

$$F_n = C_0 + n\log 2 - \log|\sin(2^n\theta)|. \tag{3.7}$$

The inequalities $0 < |\sin(2^n\theta)| \leq 1$ and (3.2) give (3.3). □

Corollary 5 (Quality criterion)**.** *Put $q_n = \log c_n/\log R_n$. Then*

$$q_n \longrightarrow 1 \quad\Longleftrightarrow\quad \log W_n = o(\log c_n), \tag{3.8}$$

and

$$\limsup_{n\to\infty}(q_n - 1) > 0 \quad\Longleftrightarrow\quad \limsup_{n\to\infty}\frac{\log W_n}{\log c_n} > 0. \tag{3.9}$$

For every fixed $\delta > 0$, the exact threshold is

$$q_n \geq 1 + \delta \quad\Longleftrightarrow\quad \log W_n + F_n \geq \frac{\delta}{1+\delta}\log c_n. \tag{3.10}$$

Proof. By (2.9), $R_n < c_n$ for all sufficiently large n. Write $\Delta_n = \log(c_n/R_n) = \log W_n + F_n$. Since $\log c_n = 2^n \log c_0$, Theorem 4 gives $F_n = o(\log c_n)$. Now

$$q_n - 1 = \frac{\Delta_n}{\log R_n},$$

which proves (3.8)–(3.9). Rearranging this identity gives (3.10). □

Proposition 6 (General Lucas form)**.** *For the Lucas V-sequence with parameters*

$$P = 2(b_0 - a_0), \qquad Q = c_0^2,$$

which satisfy $\gcd(P, Q) = 1$*, one has*

$$d_j = -\frac{1}{2} V_{2^j}(P, Q). \tag{3.11}$$

Proof. Because c_0 is odd and $\gcd(c_0, b_0 - a_0) = 1$, the parameters are coprime. The two roots of $X^2 - PX + Q$ are $\gamma = b_0 - a_0 + 2\sqrt{-a_0 b_0}$ and its conjugate. Thus

$$\frac{1}{2} V_m(P, Q) = c_0^m T_m\left(\frac{b_0 - a_0}{c_0}\right).$$

Take $m = 2^j$ and compare with (2.5). □

Corollary 7 (Dyadic negative-discriminant triples)**.** *Let $U_m = U_m(P, Q)$ be the companion sequence to the V-sequence above, and put*

$$D = P^2 - 4Q = -16 a_0 b_0.$$

For $m = 2^n$,

$$(a_{n+1}, b_{n+1}, c_{n+1}) = \left(\frac{|D| U_m^2}{4}, \frac{V_m^2}{4}, Q^m\right). \tag{3.12}$$

Moreover,

$$\lim_{n \to \infty} \frac{\log |U_{2^n}^2|}{\log Q^{2^n}} = 1. \tag{3.13}$$

Proof. The roots in the proof of Proposition 6 are $\gamma = c_0 e^{i\theta}$ and $\bar{\gamma} = c_0 e^{-i\theta}$. Hence

$$U_m = \frac{c_0^m \sin(m\theta)}{2\sqrt{a_0 b_0}}.$$

For $m = 2^n$, equations (2.3) and (2.5) give

$$a_n b_n = \frac{c_n^2 - d_n^2}{4} = a_0 b_0 U_m^2.$$

Together with $V_m/2 = -d_n$, this proves (3.12). Also,

$$\log |U_m^2| = \log Q^m + 2 \log |\sin(m\theta)| - \log(4 a_0 b_0).$$

Theorem 4 makes the last two terms $O(n)$, while $\log Q^{2^n} = 2^{n+1} \log c_0$, proving (3.13). □

Remark 8. For $n \geq 1$, both U_{2^n} and V_{2^n} are even, so (3.12) is the normalized 1/4-case of the Lucas *abc*-triples in [4, §2.2]. Bolvardizadeh later singled out the limit (3.13) as an obstacle for negative discriminants. Hajdu and Tijdeman's subsequent all-index growth theorem [8, Theorem 2.1] implies that limit for every nondegenerate non-real Lucas sequence. The proof above is retained only as a short dyadic derivation which exposes the same sine factor appearing in (2.8); no all-index novelty is claimed. Neither estimate by itself proves an unconditional quality limit.

4 The orbit of (1,8,9)

We now specialize to

$$(a_0, b_0, c_0) = (1, 8, 9), \qquad \theta = \arccos(7/9).$$

The first triples are

$$(1, 8, 9), \quad (32, 49, 81), \quad (6272, 289, 6561).$$

Since $R_0 = 6$ and $\sin\theta = 4\sqrt{2}/9$, (2.8) becomes

$$\boxed{\frac{R_n}{c_n} = \frac{3\,|\sin(2^n\theta)|}{2^{n+1}\sqrt{2}\,W_n}.} \tag{4.1}$$

There is a smaller-parameter Lucas description in this case. Let

$$A_m = 3^m T_m(1/3) = \frac{1}{2}V_m(2, 9).$$

The composition identity for Chebyshev polynomials and $T_2(1/3) = -7/9$ give

$$|d_j| = |A_{2^{j+1}}| = \frac{1}{2}\big|V_{2^{j+1}}(2, 9)\big|. \tag{4.2}$$

As noted in the introduction, this sequence identity is prior art [11].

4.1 Exact Lucas–Wieferich contributions

Let $U_m = U_m(2, 9)$, and put

$$\alpha = 1 + 2\sqrt{-2}, \qquad \beta = 1 - 2\sqrt{-2}, \qquad u = \alpha/\beta.$$

Then

$$U_m = \frac{\alpha^m - \beta^m}{\alpha - \beta}, \qquad V_m = \alpha^m + \beta^m.$$

Proposition 9 (Rank and valuation)**.** *Suppose that $p \mid d_j$, put $m = 2^{j+1}$, and let $\chi_p = \left(\frac{-2}{p}\right)$. Then $p \nmid 6$. For a prime $\mathfrak{p} \mid p$, write $\widetilde{u}$ for the image of u in $(\mathcal{O}_K/\mathfrak{p})^\times$. Then*

$$\operatorname{ord}_{(\mathcal{O}_K/\mathfrak{p})^\times}(\widetilde{u}) = 2m = 2^{j+2} \quad \text{for every prime } \mathfrak{p} \mid p \text{ in } \mathbb{Q}(\sqrt{-2}). \tag{4.3}$$

Consequently,

$$2^{j+2} \mid p - \chi_p. \tag{4.4}$$

Moreover,

$$v_p(d_j) = v_p(U_{p-\chi_p}(2, 9)). \tag{4.5}$$

In particular, $p^2 \mid d_j$ if and only if p is a Lucas–Wieferich prime for $U(2, 9)$.

Proof. Lemma 1 gives $p \nmid 18$, and d_j is odd, so $p \nmid 6$. The rank statements are standard; we include the short argument. Modulo $\mathfrak{p}$, (4.2) gives $u^m = -1$. Since m is a power of 2, the order is exactly $2m$. If p splits, this order divides $p - 1$; if p is inert, Frobenius gives $\widetilde{u}^p = \widetilde{\overline{u}} = \widetilde{u}^{-1}$, so the order divides $p + 1$. This proves (4.4), equivalently the rank criterion in Ribenboim [13, §2.13].

Sun's Lucas law of repetition [18, Theorem 3(ii)] states, under the hypotheses satisfied by $(P, Q) = (2, 9)$, that an odd prime $p \mid V_m(P, Q)$ satisfies

$$v_p(V_m) = v_p(m) + v_p\left(U_{p-(D/p)}(P, Q)\right), \qquad D = P^2 - 4Q.$$

Here $D = -32$, $(D/p) = \chi_p$, and $v_p(m) = 0$. Division by 2 does not affect the p-adic valuation, proving (4.5). $\square$

Remark 10. Sun's theorem and his analogous iterated companions $S_{k+1} = S_k^2 - 2$ already provide the per-prime valuation analysis used in Proposition 9. Our new statement is the coupling of all such valuations across the *abc*-orbit in (4.6), not the individual Lucas–Wieferich criterion.

Combining Proposition 9 with pairwise coprimality gives the exact expansion

$$\boxed{W_n = \prod_{j=0}^{n-1} \prod_{p \mid d_j} p^{\,v_p(U_{p-\chi_p}(2,9))-1}.} \tag{4.6}$$

Thus (3.8)–(3.9) are precisely statements about the aggregate weight of Lucas–Wieferich primes encountered at dyadic ranks. This is more specific than the classical one-directional implications from the *abc* conjecture to the existence of non-Wieferich primes [16, 15, 20].

4.2 What unconditional radical bounds give

There are unconditional results directly relevant to (4.2). Stewart [17, Theorem 1] proves an effective radical lower bound for Lucas and Lehmer sequences. In Stewart's notation $Q(N) = \operatorname{rad}(|N|)$, $q(m)$ is the number of square-free divisors of m, $d(m)$ is the number of positive divisors, and $|2|_2 = 1/2$. For the companion sequence his displayed inequality, after taking logarithms, is

$$\log Q(v_m) > \frac{c_S\, d(m|m|_2)(\log m)^2}{q(m) \log\log m}$$

for an effective constant $c_S > 0$ and all sufficiently large m. At $m = 2^k$, one has $q(m) = 2$, $m|m|_2 = 1$, and $d(m|m|_2) = 1$. Since $\log m = k \log 2$ and $\log\log m \asymp \log k$, this gives

$$\log \operatorname{rad}\bigl(|V_{2^k}(2,9)|\bigr) \gg \frac{k^2}{\log k}. \tag{4.7}$$

For the roots α, β above, $(\alpha+\beta)^2 = 4$, $\alpha\beta = 9$, and $\gcd(4,9) = 1$; moreover α/β is not a root of unity because $u + u^{-1} = -14/9$. These verify the hypotheses of Stewart's companion-sequence estimate. Since removing the factor 2 changes the logarithm of the radical by at most $\log 2$, summation in (2.4) gives

$$\log R_n \gg \frac{n^3}{\log n}, \qquad q_n \ll \frac{2^n \log n}{n^3}. \tag{4.8}$$

This is far below the exponential-in-n radical scale required by (3.8), but it explains precisely what presently available general theorems contribute. Primitive-divisor results alone address the appearance of new primes, whereas here every prime is already confined to one dyadic term by Lemma 1; their multiplicities are the unresolved issue; compare Bilu, Hanrot, and Voutier [3].

5 Computation

The accompanying program `check_square_lifts.py` works entirely modulo p^2. For each prime p, it iterates

$$(c,d) \longmapsto (c^2, c^2 - 2d^2) \pmod{p^2}, \qquad (c,d) = (9,-7),$$

and reports p, j exactly when $d_j \equiv 0 \pmod{p^2}$. If $p \mid d_j$ but $p^2 \nmid d_j$, the search for that prime stops: Lemma 1 proves that p cannot occur later.

Proposition 11 (Finite verification). *For every prime $p \le 10^7$, $p \neq 2, 3$, and every $0 \le j \le 50$, the modular recurrence gives $p^2 \nmid d_j$. The search tests* 664,577 *primes and reports zero square lifts.*

This finite verification is not an input to any theorem. The script contains a deterministic exact-versus-modular self-check for small indices. The command used for Proposition 11 is

```
python3 check_square_lifts.py --prime-limit 10000000 --max-j 50.
```

The first terms provide short independent checks of the recurrence, factorization, and congruence (4.4):

j	d_j	$\lvert d_j\rvert$ factored	$p \bmod 2^{j+2}$
0	-7	7	-1
1	-17	17	$+1$
2	5983	$31 \cdot 193$	$-1, +1$
3	-28545857	$2753 \cdot 10369$	$+1, +1$
4	223288285122943	$127 \cdot 19841 \cdot 88613249$	$-1, +1, +1$

Code and data availability

The source package accompanying this article contains `check_square_lifts.py`. It uses only the Python 3 standard library and generates the reported counts directly; no external data set is used. The command and all search bounds are stated above.

6 Conclusion

The exact identity (2.8) isolates the two possible sources of radical loss in every primitive opposite-parity quadratic-transfer orbit. The archimedean source is completely controlled: it contributes only $\Theta(n)$ to $\log(c_n/R_n)$. Any fixed asymptotic quality gap must therefore come from positive-power accumulation in W_n.

For the seed $(1, 8, 9)$, formula (4.6) identifies that accumulation exactly with weighted Lucas–Wieferich valuations at dyadic ranks. The finite search and the known radical bounds are consistent with sub-power growth, but neither decides it. Proving

$$\sum_{j<n} \log \frac{|V_{2^{j+1}}(2,9)|/2}{\operatorname{rad}(|V_{2^{j+1}}(2,9)|/2)} = o(2^n)$$

would prove $q_n \to 1$ for this orbit; positive linear growth on the 2^n scale along a subsequence would instead give a fixed quality gap. Both remain open.

AI-use statement

Anthropic Claude Fable 5 and OpenAI Codex (GPT-5-based service model), using the service versions available on July 26, 2026, were used for idea exploration, candidate proofs, literature classification and retrieval, coding assistance, drafting, and adversarial cross-review during manuscript development. Anthropic Claude Opus 5 was used for the final independent pull-request review. These systems were used to accelerate exploration and to expose errors through independent cross-checking. Mathematical claims were line-checked in separate review passes, bibliographic claims were checked against cited sources, and the computation was independently reproduced. The AI systems are not authors. The named author directed the project and accepts responsibility for the originality, validity, references, code, and final content.

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